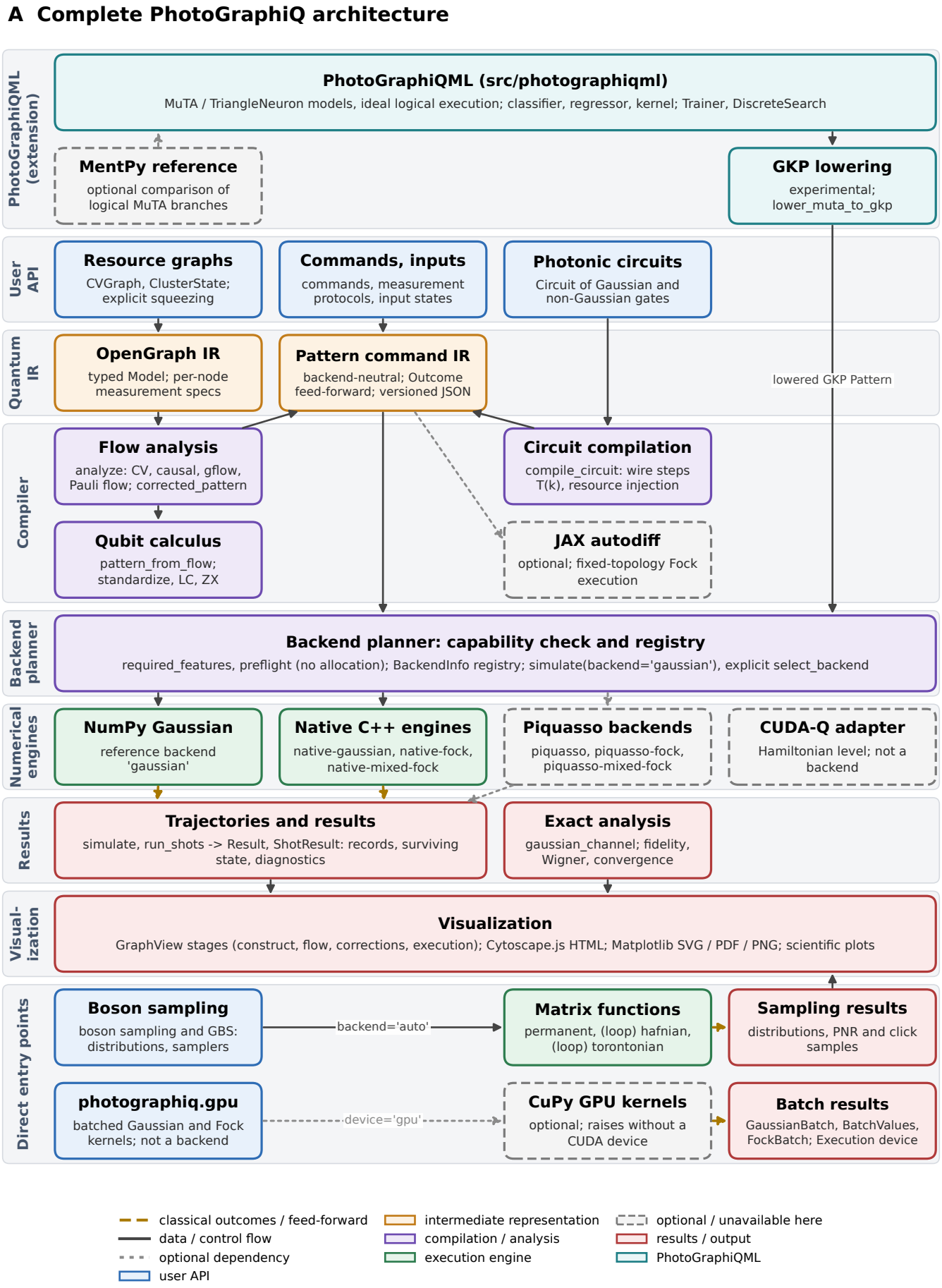
A Complete PhotoGraphiQ architecture
PhotoGraphiQML (extension)
PhotoGraphiQML (src/photographiqml)
MuTA / TriangleNeuron models, ideal logical execution; classifier, regressor, kernel; Trainer, DiscreteSearch
MentPy reference
optional comparison of logical MuTA branches
GKP lowering
experimental; lower_muta_to_gkp
User API
Resource graphs
CVGraph, ClusterState; explicit squeezing
Commands, inputs
commands, measurement protocols, input states
Photonic circuits
Circuit of Gaussian and non-Gaussian gates
Quantum IR
OpenGraph IR
typed Model; per-node measurement specs
Pattern command IR
backend-neutral; Outcome feed-forward; versioned JSON
lowered GKP Pattern
Compiler
Flow analysis
analyze: CV, causal, gflow, Pauli flow; corrected_pattern
Circuit compilation
compile_circuit: wire steps T(k), resource injection
Qubit calculus
pattern_from_flow; standardize, LC, ZX
JAX autodiff
optional; fixed-topology Fock execution
Backend planner
Backend planner: capability check and registry
required_features, preflight (no allocation); BackendInfo registry; simulate(backend='gaussian'), explicit select_backend
Numerical engines
NumPy Gaussian
reference backend 'gaussian'
Native C++ engines
native-gaussian, native-fock, native-mixed-fock
Piquasso backends
piquasso, piquasso-fock, piquasso-mixed-fock
CUDA-Q adapter
Hamiltonian level; not a backend
Results
Trajectories and results
simulate, run_shots -> Result, ShotResult: records, surviving state, diagnostics
Exact analysis
gaussian_channel; fidelity, Wigner, convergence
Visual-ization
Visualization
GraphView stages (construct, flow, corrections, execution); Cytoscape.js HTML; Matplotlib SVG / PDF / PNG; scientific plots
Direct entry points
Boson sampling
boson sampling and GBS: distributions, samplers
backend='auto'
Matrix functions
permanent, (loop) hafnian, (loop) torontonian
Sampling results
distributions, PNR and click samples
photographiq.gpu
batched Gaussian and Fock kernels; not a backend
device='gpu'
CuPy GPU kernels
optional; raises without a CUDA device
Batch results
GaussianBatch, BatchValues, FockBatch; Execution device
classical outcomes / feed-forward
data / control flow
optional dependency
user API
intermediate representation
compilation / analysis
execution engine
optional / unavailable here
results / output
PhotoGraphiQML